ECON 185 Santa Clara Stop-and-Frisk Audit

Ariel Tran

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Abstract

This case study explores the possible of racial profiling in stops and searches. Using the 2019 RIPA dataset for Santa Clara County (44,306 recorded stops), this report compares descriptive statistics and logistic regression output to compared how stopped individual that are Black or Hispanic/Latino compare to other racial groups. For stopped individuals of these minority groups, the analysis found higher search rates, lower hit rates, and a statistically significant lower chance of hits. These results suggest policing behavior that is biased against people that were perceived as Black or Hispanic/Latino.

1 Context

I chose 2019 RIPA data from Santa Clara County. This data set was created and released because of the Racial and Identity Profiling Act of 2015 that aims to combat unlawful racial and identity-based profiling. There are 44,306 total recorded stops in Santa Clara County during 2019.

2 Summary Statistics

To analyze how the perceived race of a person stopped may relate to stops, the count and percentages are broken down (Table 1 and Table 2). The racial groups with the greatest proportion of stops includes Hispanic/Latino (49.3%), White (22.9%), and Asian (11.7%).

Table 1: Breakdown of Total Stops Across Perceived Racial Groups

Perceived Race by Officer	**Stops (Count and %)**
Hispanic/Latino	21863 (49.3%)
White	10158 (22.9%)
Asian	5200 (11.7%)
Black	4281 (9.7%)
Middle Eastern / South Asian	1640 (3.7%)
Multiracial	745 (1.7%)
Pacific Islander	354 (0.8%)
Native American	65 (0.1%)

The most common reasons for a stop are traffic violations and reasonable suspicion. These two reasons for stops account for 91.2% of all recorded stops in Santa Clara County during 2019. When breaking down reasons for stops within all stops for a racial group, the top three racial groups vary across traffic violations, reasonable suspicions, and consensual encounters resulting in a search.

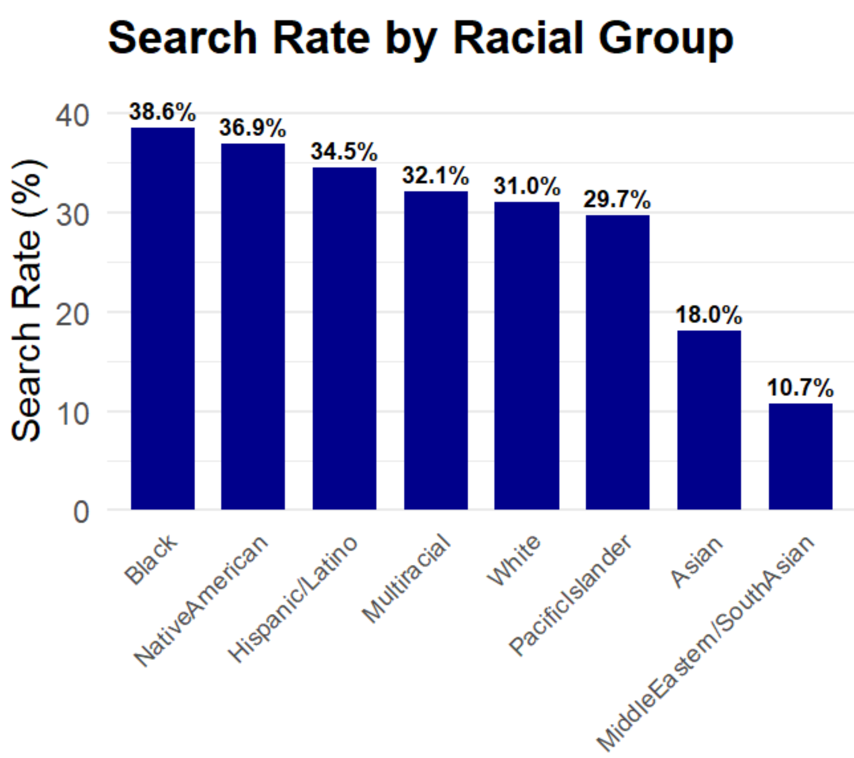
Table 2: Breakdown of Reasons for Stops by Perceived Race

Perceived Race by Officer	**Traffic Violation (%)**	**Reasonable Suspicion (%)**	**Searched (%)**
Hispanic/Latino	58.2%	31.5%	4.5%
White	55.3%	36.5%	4.9%
Asian	75.2%	18.7%	3.6%
Black	46.8%	45.4%	4.0%
Middle Eastern / South Asian	81.6%	15.9%	1.5%
Multiracial	59.6%	30.2%	3.9%
Pacific Islander	55.9%	34.7%	4.0%
Native American	35.4%	47.4%	9.2%

Search rates are calculated by investigating the proportion of searches out of all stop within a racial group. For this particular analysis, searches included searches for the person and their property. If both types of searches occurred in the same stop, it was counted as one instance.

The search rate varies greatly (10.7 - 38.6%) across all racial groups. Putting aside racial groups that account for less than 4% of the total stops, the top three search rates by racial group includes Black (38.6%), Hispanic/Latino (34.5%), and White (31.0%).

Figure 1:



The disparity in search rates between racial groups does not prove or disprove biased policing. Although Black and Hispanic/Latino stops were searched at some of the highest rates is consistent with the racial profiling, these numbers do not establish that biased policing is sole or main cause of disproportional search rates. Disproportionate search rates do not show if those differences are due to biased policing or different hit rates across racial groups. Including hit rates (hits/searches) can suggest if officers are searching certain races on a lower or higher suspicion threshold. Therefor, deeper analysis is necessary to prove stop-and-frisks in Santa Clara during 2019.

3 The Veil of Darkness (VOD) Test

Performing the veil of darkness test investigates if lower racial visibility at dark hours correlates less stops of minorities, which would then be consistent with racial profiling. To explore this theory, I utilized a generalized linear model (glm) in order to perform logistic regression. The coefficients in this method represents the change in probability of the stopped individual being Black/Hispanic/Latino (n = 26144) or another perceived race (n = 18162) when it's dark out.

To measure darkness in a simple manner, I defined it as 7:00 PM until 5:00 AM. This measurement of darkness accounts for most of the darkness across the varying dawn and dusk windows throughout the twelve months of a calendar year.

Another critical component of this analysis requires controlling for time. It was imperative that the regression not pick up on how driving and policing patterns likely differ during daylight and darkness. To control for time I put the continuous twenty-four hour formatted time of the stop into fifteen minute bins and factored it. Interpreting time in quarter chunks of the hour makes sure that the sixty minutes counting based is not misinterpreted as part of the one hundred based counting format. Factoring this variables ensures that the model is comparing similar fifteen minute chunks across days instead of against other fifteen minute chunks throughout the progression of a singular day. Holding time constant better isolates any statistical relationship between visibility and potentially racially motivated stops.

Table 3: Logistic Regression of Stopped Person Being Black/Hispanic/Latino During Daylight vs. Dark Hours

Variable	Coefficient	Std. Error	Z-value	P-value
Dark Hours (7PM-5AM)	0.1784	0.1782	1.001	0.3168
Daylight	0.2018	0.1979	1.020	0.3079

The probability of the stopped driver being Black/Hispanic/Latino when the sun is down (55.0%) is slightly lower than when the sun is up (59.4%,). Due to the high p-value, the results of this regression are statistically insignificant. Therefore these coefficients are likely due to chance rather than a reflection of any possible relationship between visibility and being a stopped minority.

4 Becker's Outcome Test: Hit Rates

To explore the relationship between race and hit rates, I made a logistic regression model (glm). Hit rates are defined as the total instances of contraband found out of the total searched within a racial group. I compared Black/Hispanic/Latino (n = 26144) to White (n = 10158) , so that White would serve as the theoretically more "fair" reference group and control.

- White: Conditional on already having their person and/or property searched, White drivers have a hit rate of 37.0%.
- Black/Hispanic/Latino: Conditional on already having their person and/or property searched, Black/Hispanic/Latino drivers have a hit rate 3.2% lower hit of 33.8%.

Table 4: Logistic Regression of Contraband Being Found During a Search for Black/Hispanic/Latino vs White

Perceived Race	Coefficient	Std. Error	Z-value	P-value
Black/Hispanic/Latino	-0.1412	0.0430	-3.286	¡ 2e-16
White	-0.53017	0.0369	-14.370	0.00102

The results of the logistic regression model that White drivers are more likely to have contraband are consistent with their higher hit rate percentages (See Table 5).

Table 5: Hit Rates Across Black/Hispanic/Latino and White

Perceived Race	Search Rate	Hit Rate
Black/Hispanic/Latino	35.2%	43.4%
White	31.0%	46.8%

5 Conclusion

The Veil of Darkness test has results that were inconsistent with the theory racial bias in policing. The statistically insignificant results means that this analysis has insufficient evidence to make a conclusion about racial profiling in Santa Clara County's policing.

Comparing the search and hit rates between Black/Hispanic/Latino and White suggest policing bias against Black/Hispanic/Latino. Even with statistically significant regression outputs that are consistent with the descriptive statistics, the differences are relatively small between these two groups.

- *Black/Hispanic/Latino*: higher search rates, lower hit rates, and statistically significant lower chances of a hit during a search. This suggests policing on more flimsy reasoning and biased policing.
- *White:* lower search rates and higher hit rates, and statistically significant higher chances of a hit during a search. This suggest a higher suspicion threshold and policing bias in favor of White individuals.

Because this is a case study on one county and year, larger scale analysis will help with making a statistically stronger conclusion. Overall, this case study has multiple limitations and minor evidence suggesting the presence of stops-and-frisks.